



Summer Internship at AICW, Edunet Foundation

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Internal Guide

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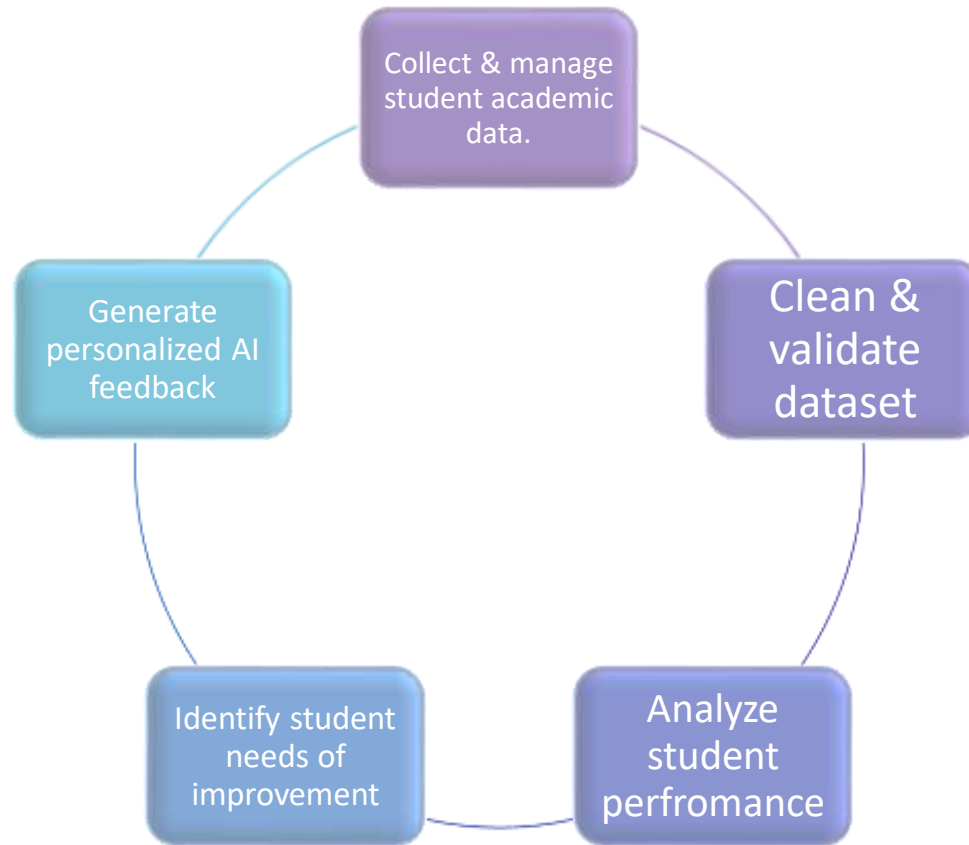


Company's Introduction:

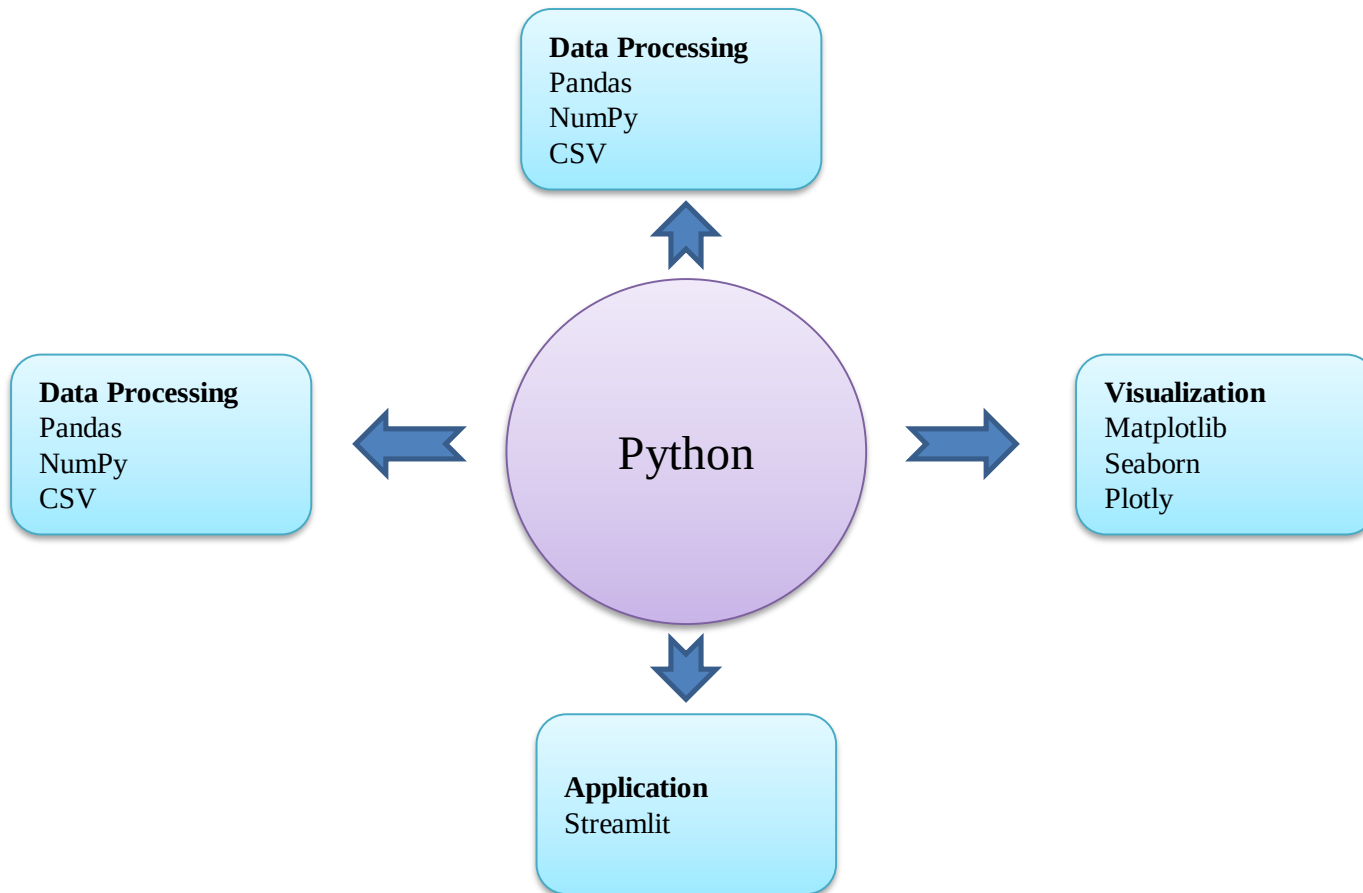
- Edunet Foundation is a non-profit organization working in the areas of skilling, livelihood enhancement, and STEM education.
- AICW is a national initiative by Edunet Foundation that empowers Non-STEM women with future-ready AI skills, mentorship, and certification, delivered in collaboration with Microsoft, SAP and LinkedIn, with support from the Ministry of Skill Development & Entrepreneurship.
- **Official designation:** AI Intern
- **Duration:** 4 weeks (06-07-2026 to 31-07-2026)
- **Mode:** Offline



Objectives of Internship's Project:



Technology Stack:



Project Overview & Problem Statement:

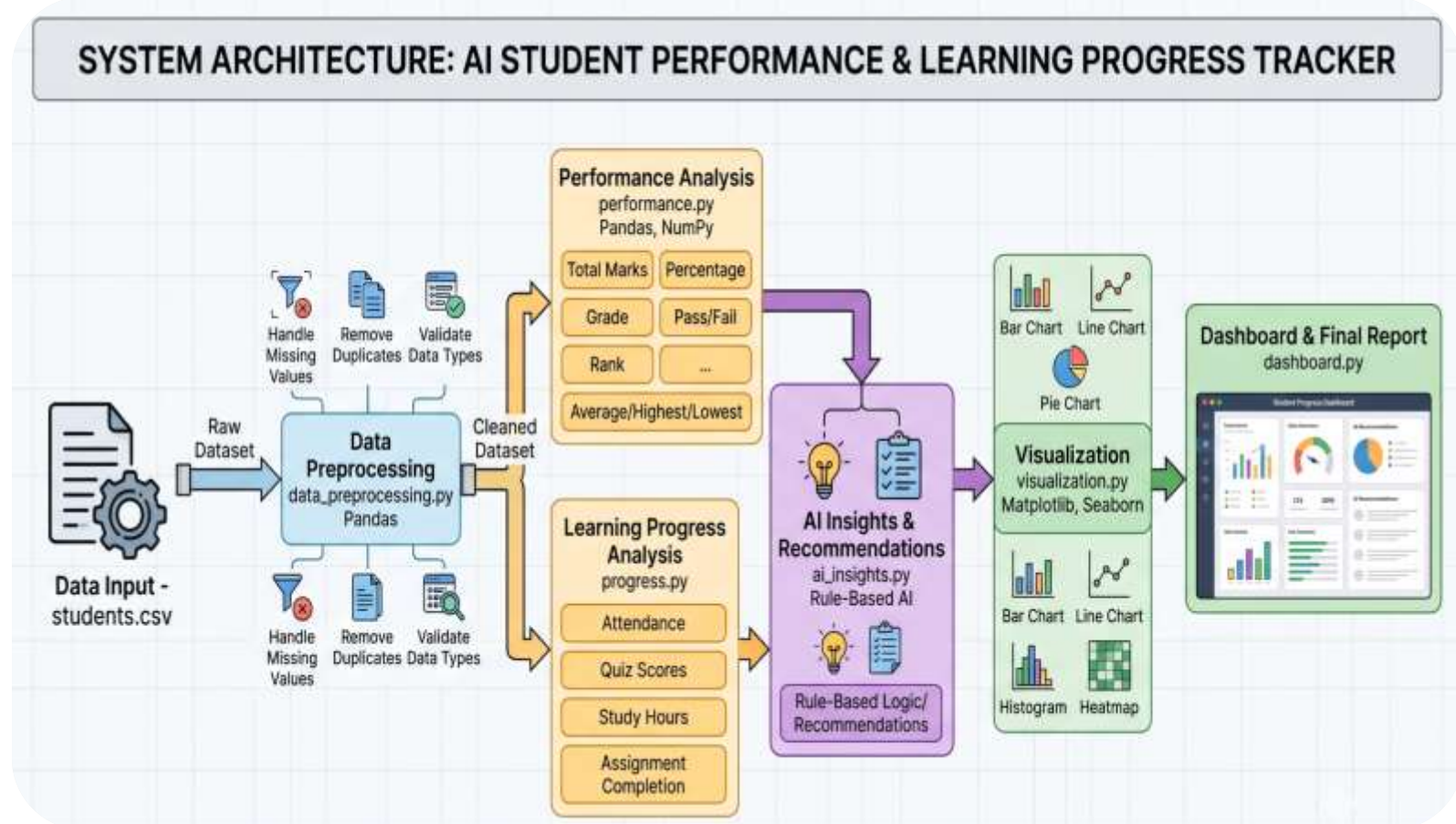
Project Overview

- Project **The AI Student Performance & Learning Progress Tracker** is a Python-based data analytics and AI application developed to monitor and analyze students' academic performance and learning activities. The system processes student data, performs performance and progress analysis, generates personalized AI-based feedback, and presents the results through an interactive dashboard.
- The project uses **Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Streamlit, and Google Gemini API**. It combines rule-based analysis with Gemini-powered feedback to provide meaningful insights into students' strengths and areas for improvement.

Problem Statement

- Traditional student performance evaluation mainly focuses on examination marks and may not provide a complete picture of a student's learning progress. Factors such as **attendance, quiz performance, study hours, and previous academic performance** also need to be considered.
- Manually analyzing these parameters for a large number of students can be time-consuming and makes it difficult to identify students who require additional support. There is also a need for personalized and easily understandable feedback rather than only numerical results.

System Architecture & Workflow:

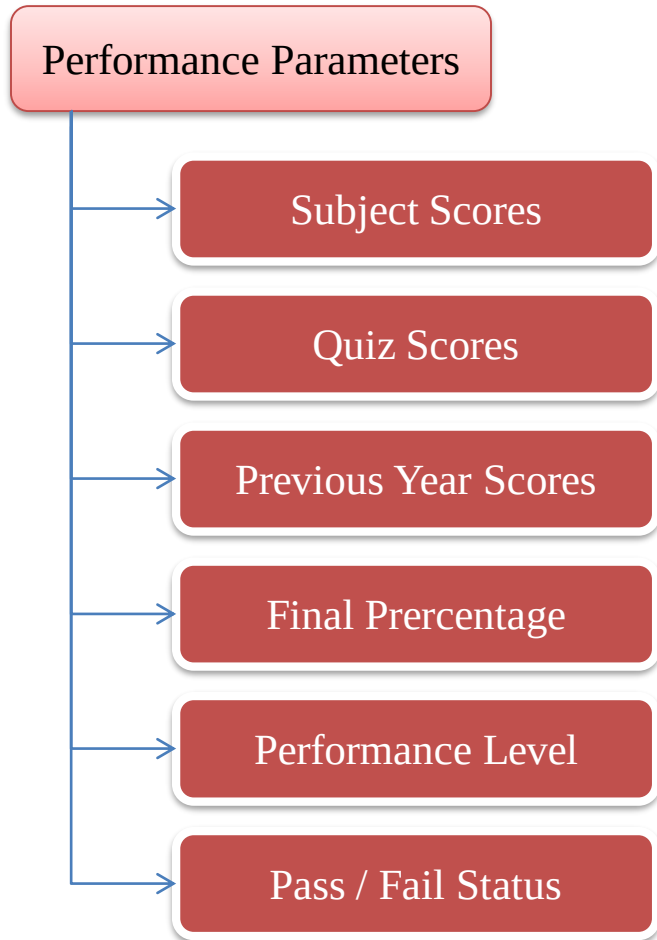


Dataset & Data Preprocessing:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Student_ID	Age	Gender	Study_Hou	Attendance	Internet_A	Extracurric	Math_Scor	Science_S	English_Sc	Quiz_Scor	Previous_Y	Final_Perc	Performan	Pass_Fail
S0001	15	Male	1	65	No	No	40	39	72	48	81	50.33	Average	Pass
S0002	19	Female	1.6	58	No	Yes	80	44	35	53	47	53	Average	Pass
S0003	14	Female	3.6	64	Yes	Yes	83	73	59	72	58	71.67	Good	Pass
S0004	18	Female	5.5	68	Yes	No	68	48	77	63	54	64.33	Average	Pass
S0005	14	Male	5	80	Yes	No	41	46	36	41	66	41	Poor	Fail
S0006	15	Male	5.2	82	No	Yes	88	70	46	66	60	68	Good	Pass
S0007	16	Male	0.8	78	No	No	75	69	64	67	84	69.33	Good	Pass
S0008	18	Female	2	81	No	Yes	93	37	48	60	55	59.33	Average	Pass
S0009	18	Female	3.5	61	No	No	55	39	83	60	71	59	Average	Pass
S0010	18	Male	5.7	97	Yes	No	72	73	73	75	61	72.67	Good	Pass
S0011	14	Female	3.6	88	Yes	No	95	85	69	83	41	83	Excellent	Pass
S0012	18	Female	5.7	74	Yes	No	54	38	59	49	81	50.33	Average	Pass
S0013	19	Male	1.5	64	Yes	Yes	35	95	37	58	85	55.67	Average	Pass
S0014	17	Female	5.5	76	No	Yes	79	82	100	85	71	87	Excellent	Pass
S0015	15	Female	4.2	60	No	Yes	55	57	80	63	72	64	Average	Pass
S0016	17	Male	5	67	No	No	68	40	81	63	72	63	Average	Pass
S0017	16	Male	4	66	No	Yes	49	59	96	68	63	68	Good	Pass
S0018	14	Female	5	89	No	Yes	71	40	59	58	82	56.57	Average	Pass
S0019	14	Female	1.9	100	No	No	45	85	74	69	61	68	Good	Pass
S0020	19	Male	3.6	66	Yes	Yes	75	78	59	69	78	70.67	Good	Pass
S0021	15	Female	4.7	65	No	Yes	48	80	70	64	46	65.33	Average	Pass
S0022	15	Male	2.8	83	No	Yes	100	96	99	98	86	98.33	Excellent	Pass
S0023	19	Female	5.3	72	No	No	72	40	63	59	76	58.33	Average	Pass
S0024	14	Female	4.6	54	Yes	No	58	37	79	58	94	58	Average	Pass
S0025	17	Male	0.9	98	Yes	No	61	53	38	51	48	50.67	Average	Pass

- Dataset contains **5,000 student records**.
- Loaded student dataset using Pandas.
- Inspected dataset structure using **info()**, **shape()**, and **describe()**.
- Validated score ranges between 0 and 100.

Student Performance & Learning Progress Analysis:



Method Used	Calculations
mean()	Average Percentage
max()	Highest Percentage
min()	Lowest Percentage
value_counts()	Pass/Fail & Performance Distribution
idxmax()	Top Student

Student Performance & Learning Progress Analysis (Conti..):

```
1 # -----
2 # Attendance Analysis
3 # -----
4 def analyze_attendance(student):
5
6     if student["Attendance_Percentage"] >= 90:
7         return "Excellent"
8
9     elif student["Attendance_Percentage"] >= 75:
10        return "Good"
11
12    else:
13        return "Poor"
```



Attendance Count

- The system classifies attendance, quiz scores, and study hours and combines these results to determine the student's overall progress.

Quiz Analysis



```
1 # -----
2 # Quiz Analysis
3 # -----
4 def analyze_quiz(student):
5
6     if student["Quiz_Score"] >= 80:
7         return "Excellent"
8
9     elif student["Quiz_Score"] >= 50:
10        return "Average"
11
12    else:
13        return "Poor"
```

Student Performance & Learning Progress Analysis (Conti..):

```
1 # -----
2 # Overall Progress
3 # -----
4 def analyze_progress(student):
5
6     attendance = analyze_attendance(student)
7     quiz = analyze_quiz(student)
8     study = analyze_study_hours(student)
9
10    if (
11        attendance == "Excellent"
12        and quiz == "Excellent"
13        and study == "Excellent"
14    ):
15        overall = "Excellent"
16
17    elif (
18        attendance == "Poor"
19        or quiz == "Poor"
20        or study == "Poor"
21    ):
22        overall = "Needs Improvement"
23
24    else:
25        overall = "Good"
26
27    return {
28        "Attendance_Status": attendance,
29        "Quiz_Status": quiz,
30        "Study_Hours_Status": study,
31        "Overall_Progress": overall
32    }
33
```



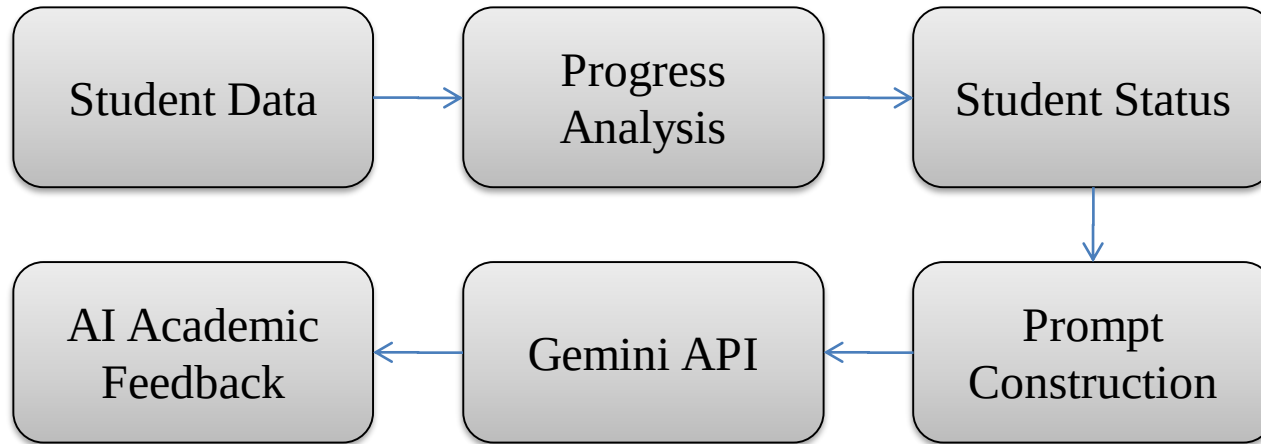
Overall Progress

Study Hours



```
1 # -----
2 # Study Hours Analysis
3 # -----
4 def analyze_study_hours(student):
5
6     if student["Study_Hours_Per_Day"] >= 4:
7         return "Excellent"
8
9     elif student["Study_Hours_Per_Day"] >= 2:
10        return "Good"
11
12    else:
13        return "Poor"
```

AI-Based Insights & Personalized Recommendations:



- The rule-based module first converts raw values into meaningful statuses. These results are then provided to Gemini through a structured prompt. Gemini acts as an academic mentor and generates simple, practical, and personalized feedback.

Data Visualization & Interactive Dashboard:



Figure 1: Dashboard showing comparison of Study hrs. vs final % using line scatter analysis & average student performance using bar graph.

Data Visualization & Interactive Dashboard (Conti.):



Figure 2: Dashboard showing Pass / Fail distribution using pie chart & attendance distribution using histogram.

Data Visualization & Interactive Dashboard (Conti.):



Figure 3: Dashboard showing Student performance correlation using heatmap.

Data Visualization & Interactive Dashboard (Conti.):



Figure 4: Student dashboard.

Challenges and Learnings:

Challenges:

- Raw dataset required validation.
- Large dataset difficult to analyze manually.
- Academic scores alone were insufficient.
- Rule-based output was limited.
- Static analysis was difficult to explore.
- Large amount of information.

Learnings:

- Python programming.
- Automated Calculations using python.
- API Integration.
- Using of charts & KPIs.
- Data processing.
- Data visualization.

Future Scope:

- Machine Learning-based performance prediction
- Student login system
- Advanced Streamlit dashboard
- PDF report generation Email notifications
- Database integration
- More detailed student progress tracking
- Historical progress comparison

Conclusion:

- The project successfully combines data preprocessing, academic performance analysis, learning-progress tracking, AI-generated feedback, and data visualization into a single application. Through this project, we gained practical experience in Python, data analytics, visualization, API integration, and Streamlit application development. The system demonstrates how student data can be converted into meaningful insights and personalized learning recommendations.

References:

- [1] Python Official Documentation. Available: <https://docs.python.org/3/>
- [2] Pandas Official Documentation. Available: <https://pandas.pydata.org/docs/>
- [3] NumPy Official Documentation. Available: <https://numpy.org/doc/>
- [4] Matplot Lib Official Documentation. Available: <https://matplotlib.org/stable/>
- [5] Seaborn Official Documentation. Available: <https://seaborn.pydata.org/>
- [6] Kaggle Official Documentation. Available: <https://www.kaggle.com/>

THANK YOU
